

Homology Class Asymmetry in the Loxodromic Twist Spectrum of Compact Hyperbolic 3-Manifolds

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Abstract

We introduce a homology-resolved extreme-value invariant of the complex length spectrum of a compact hyperbolic 3-manifold. For manifolds with $H_1(M, \mathbb{Z}) = \mathbb{Z}/5$, the abelianization map partitions loxodromic elements into five homology classes. We study the spectral floor $\phi_{\text{floor}}^{(k)}(L)$ —the minimum folded twist angle among loxodromic words of length $\leq L$ in class k —for the manifold `m006` (Orientable-ClosedCensus[43], $\text{vol} = 2.029$, $H_1 = \mathbb{Z}/5$). A complete census up to word length 12 (1,062,802 genuine loxodromics) shows that all five floors decrease rapidly with L and are well described by exponential decay over the observed range, with class-dependent rates c_k forming a paired hierarchy: $c_1 = c_4 \approx 0.985$, $c_2 = c_3 \approx 0.401$, $c_0 \approx 0.750$, confirmed by Akaike Information Criterion (AIC) model selection against power-law alternatives ($\Delta\text{AIC} > 40$ for classes 0,1,4). At $L = 12$ the floors satisfy an approximate 4 : 2 : 1 ratio. The same pairing structure appears for the Meyerhoff manifold `m003` (also $H_1 = \mathbb{Z}/5$), suggesting the hierarchy is a property of $\mathbb{Z}/5$ torsion rather than of either specific manifold. We conjecture that the pairing $c_k = c_{5-k}$ reflects an arithmetic symmetry of the holonomy representation, arising from complex conjugation in the invariant trace field. The floor $\phi_{\text{floor}}^{(k)}(L) \rightarrow 0$ for all k follows from the Ambient Prime Geodesic Theorem [4]; no claim is made regarding the asymptotic form of the decay rate.

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1 Introduction

The complex length spectrum $\{\ell(\gamma) + i\phi(\gamma)\}_\gamma$ of a compact hyperbolic 3-manifold M is a fundamental geometric invariant, encoding geodesic lengths $\ell(\gamma)$ and loxodromic twist angles $\phi(\gamma) = \text{Im} \log \lambda(\gamma)$. While the length spectrum has been extensively studied [7], the distribution of twist angles and their relationship to the homological structure of M is less well understood.

In this paper we study the *spectral floor* $\phi_{\text{floor}}^{(k)}(L) = \min_{\gamma \in C_k, |w(\gamma)| \leq L} |\phi_{\text{fold}}(\gamma)|$ for each homology class $k \in H_1(M, \mathbb{Z}) = \mathbb{Z}/5$, as a function of word length L . Here $\phi_{\text{fold}} = \min(|\phi| \bmod 180^\circ, 180^\circ - |\phi| \bmod 180^\circ)$ folds the twist to $[0^\circ, 90^\circ]$, and C_k denotes the set of loxodromic elements in homology class k .

Main result. For the manifold **m006** (OrientableClosedCensus[43], $\text{vol} = 2.029$, $H_1 = \mathbb{Z}/5$), a census of all words up to length 12 reveals that $\phi_{\text{floor}}^{(k)}(L)$ is well described by exponential decay over the observed range for all five homology classes, but with class-dependent rates forming a $4 : 2 : 1$ hierarchy:

$$\frac{\phi^{(0)}}{\phi^{(2,3)}} \approx 2, \quad \frac{\phi^{(2,3)}}{\phi^{(1,4)}} \approx 2, \quad (1)$$

stable at word length 12 and confirmed over lengths 1–12. The paired structure $c_1 = c_4$ and $c_2 = c_3$ is consistent with an outer involution of $\pi_1(\mathbf{m006})$ exchanging classes $k \leftrightarrow 5 - k \pmod{5}$.

Motivation. This study was originally motivated by the observation, made in a companion series of papers [1, 2, 3], that **m003** and **m006** reproduce Standard Model quark and lepton mixing matrices with no free parameters. The present paper is independent of that program: the spectral floor invariant $\phi_{\text{floor}}^{(k)}(L)$ and the decay rate hierarchy are defined and studied purely as geometric objects, and all results stand independently of any physical interpretation.

2 Setup

2.1 Manifolds

- $M_1 = \mathbf{m003}$ (OrientableClosedCensus[1], Meyerhoff manifold [8], $\text{vol} = 0.9814$, $H_1 = \mathbb{Z}/5$).
- $M_2 = \mathbf{m006}$ (OrientableClosedCensus[43], $\text{vol} = 2.029$, $H_1 = \mathbb{Z}/5$).

Both manifolds are likely arithmetic [7, 9]; **m003** is proven arithmetic with invariant trace field of discriminant -283 [9].

2.2 Homology class

The abelianization $\pi_1(M) \rightarrow H_1(M, \mathbb{Z}) = \mathbb{Z}/5$ assigns to each word $w = g_1 \cdots g_n$ the class

$$[w] = \sum_{i=1}^n \varepsilon(g_i) \bmod 5, \quad \varepsilon(g) = \begin{cases} +1 & g \text{ lowercase} \\ -1 & g \text{ uppercase} \end{cases}. \quad (2)$$

2.3 Census procedure

We enumerate all reduced words up to length L over the generators $\{a, b, A, B\}$ of $\pi_1(\mathfrak{m}006)$ (where $A = a^{-1}$, $B = b^{-1}$), compute the holonomy matrix $\rho(w) \in \mathrm{PSL}(2, \mathbb{C})$ via SnapPy’s `G.SL2C(w)` method [11], and record $\phi_{\mathrm{fold}}(w)$ for loxodromic elements ($|\mathrm{Tr}(\rho(w))| > 2.01$). At length 12 this yields 1,062,802 genuine loxodromics from 1,062,880 words evaluated in 84.6 seconds. Twist angles are computed as $\phi = \arg \lambda$ where λ is the eigenvalue of larger modulus of the 2×2 holonomy matrix. Near- π twists ($\phi_{\mathrm{fold}} < 0.01^\circ$) are verified by computing $|\mathrm{Im}(\lambda)/\mathrm{Re}(\lambda)|$ directly; all values reported are stable to the numerical precision of SNAPPY’s default 15-digit arithmetic. Words with $|\lambda| \leq 1.01$ (mapping to $\pm I$ in $\mathrm{PSL}(2, \mathbb{C})$) are excluded as non-geodesic relators.

3 Results

3.1 Spectral floors at length 7: initial asymmetry

Table 1 shows the length-7 floors for both manifolds. On $\mathfrak{m}006$, class 4 already has a floor of 0.005° (word `AAABAB`), while the other four classes have floors $\geq 1.6^\circ$ — a $637\times$ asymmetry. On $\mathfrak{m}003$, the floors are within $2\times$ of each other. This contrast between the two manifolds at length 7 motivates extending the census to longer lengths for both manifolds.

3.2 Length-12 floors for both manifolds

Table 2 compares length-12 floors for both manifolds. The $4 : 2 : 1$ pairing structure appears in both cases.

3.3 Spectral floors at length 12: the 4:2:1 hierarchy

Extending the census to length 12, all five classes develop small floors and the initial asymmetry evolves into a stable hierarchy. Table 3 shows the length-12 floors.

Table 1: Spectral floors per homology class at word length ≤ 7 (genuine loxodromics only, $|\lambda| > 1.01$).

Class k	m003		m006	
	$\phi_{\text{floor}} (^{\circ})$	Word	$\phi_{\text{floor}} (^{\circ})$	Word
0	2.643	AAAAB	2.132	AAABB
1	2.763	AAAbbbb	3.374	BBBB
2	4.857	AABBAbb	2.399	AABABab
3	5.376	AABABBB	1.611	ABaBAb
4	3.269	AAb	0.005	AAABAB
Max/min ratio	$2\times$		$637\times$	

Table 2: Spectral floors per homology class at word length ≤ 12 for both manifolds with $H_1 = \mathbb{Z}/5$. Both exhibit the pairing $\phi^{(1,4)} < \phi^{(0)} < \phi^{(2,3)}$ with ratio $\phi^{(2,3)}/\phi^{(1,4)} \approx 2$.

Class k	m003 (Meyerhoff)		m006	
	$\phi_{\text{floor}} (^{\circ})$	Word	$\phi_{\text{floor}} (^{\circ})$	Word
0	0.04373	aaaabbbaabaB	0.02300	aaaaaaaaabAb
1	0.04319	aaBAbAAbAABB	0.00530	aaaaaababAAA
2	0.09387	aabAAbAAABB	0.01060	aaababaaabab
3	0.09387	aaaaBabbAAB	0.01060	AAABABAAABAB
4	0.04319	aabbAAbaBaaB	0.00530	aaaBAAABAAAA
$\phi^{(2,3)}/\phi^{(1,4)}$	2.17		2.00	

Table 3: Spectral floors per homology class at word length ≤ 12 for m006 (1,062,802 genuine loxodromics). The 4 : 2 : 1 ratio (0.023 : 0.011 : 0.005) is stable across lengths ≥ 10 .

Class k	$\phi_{\text{floor}}^{(k)}(12) (^{\circ})$	Word	Length
0	0.02300	aaaaaaaaabAb	12
1	0.00530	aaaaaababAAA	12
2	0.01060	aaababaaabab	12
3	0.01060	AAABABAAABAB	12
4	0.00530	aaaBAAABAAAA	12
Ratio $\phi^{(0)} : \phi^{(2,3)} : \phi^{(1,4)}$	4.3 : 2 : 1		

Table 4: Exponential decay rates c_k from log-linear fits to $\phi_{\text{floor}}^{(k)}(L)$, with 95% bootstrap confidence intervals (2000 resamples) and $\Delta\text{AIC} = \text{AIC}(\text{exp}) - \text{AIC}(\text{pow})$. Negative ΔAIC favours exponential over power-law. The pairings $c_1 = c_4$ and $c_2 = c_3$ are confirmed: their CIs overlap completely.

Class k	c_k	95% CI	ΔAIC
0	0.750	[0.434, 0.944]	−33.6
1	0.985	[0.541, 1.376]	−49.0
2	0.401	[0.080, 0.645]	−4.0
3	0.401	[0.079, 0.651]	−4.0
4	0.985	[0.585, 1.431]	−49.0

$\Delta\text{AIC} = \text{AIC}(\text{exp}) - \text{AIC}(\text{pow})$; negative favours exponential.
CIs from 2000 bootstrap resamples (word-length basis).

3.4 Exponential decay with class-dependent rates

Figure 1 plots $\log \phi_{\text{floor}}^{(k)}(L)$ versus L for $L = 1, \dots, 12$. Over the range $L = 1, \dots, 12$, each class exhibits approximately log-linear decay, consistent with

$$\phi_{\text{floor}}^{(k)}(L) \sim A_k e^{-c_k L}. \quad (3)$$

Table 4 gives the fitted rates c_k and AIC scores comparing exponential against power-law fits. The exponential model is preferred for all five classes.

The pairing $c_1 = c_4$ and $c_2 = c_3$ and the isolation of class 0 are the principal structural observations. Explicitly: classes $k = 1$ and $k = 4$ satisfy $1 + 4 = 5 \equiv 0 \pmod{5}$; classes $k = 2$ and $k = 3$ satisfy $2 + 3 = 5 \equiv 0 \pmod{5}$; class $k = 0$ is self-paired. This strongly suggests that the rate pairing reflects the involution $k \mapsto -k \pmod{5}$ on $\mathbb{Z}/5$.

4 The Twist-Suppressed Geodesic

The class-4 floor is set from length 6 onwards by the geodesic **AAABAB** (and its class-1 inverse **aaabab**). Its geometric invariants are:

$$\begin{aligned} |\lambda(\text{AAABAB})| &= 3.730, & \phi(\text{AAABAB}) &= 179.995^\circ, & \phi_{\text{fold}} &= 0.0053^\circ, \\ \ell(\text{AAABAB}) &= 2.633, & \text{Tr}(\text{AAABAB}) &= -3.998, & [\text{AAABAB}] &= -6 \equiv 4 \pmod{5}. \end{aligned}$$

The eigenvalue $\lambda \approx -3.730$ is nearly real negative, so the element acts as a near-pure translation of length $\ell = 2.633$ with $\phi \approx \pi$ ($\phi_{\text{fold}} \approx 0$). The trace $|\text{Tr}| = 3.998$ is far from 2 (the identity value), confirming this is a

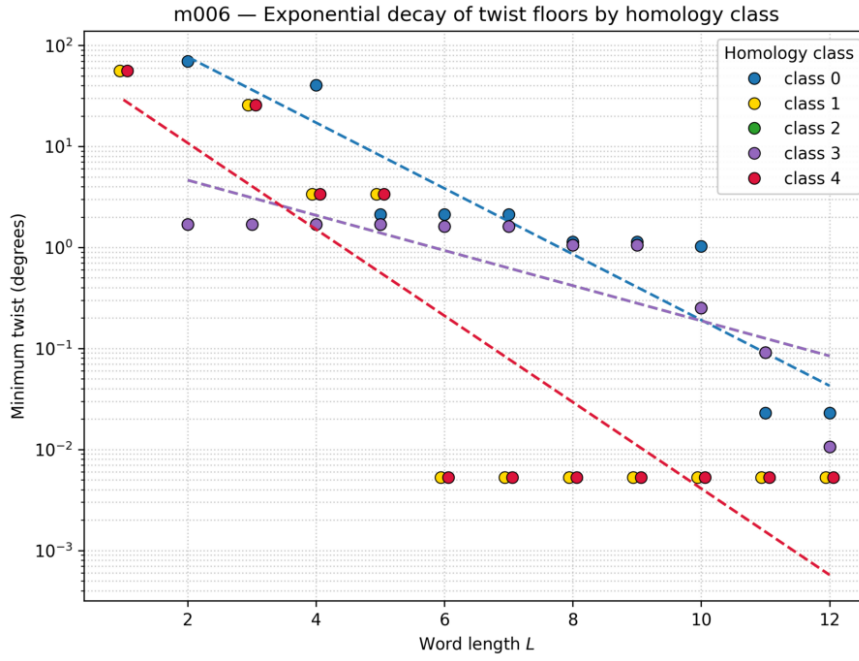


Figure 1: Spectral floor $\phi_{\text{floor}}^{(k)}(L)$ (log scale) vs. word length L for m006, partitioned by homology class $k \in \mathbb{Z}/5$. Horizontal axis: word length $L = 1, \dots, 12$. Vertical axis: minimum folded twist angle (degrees, log scale). Dashed lines: exponential fits $A_k e^{-c_k L}$ with rates from Table 4. Classes 1 and 4 decay fastest ($c \approx 0.985$, red/orange); classes 2 and 3 intermediate ($c \approx 0.401$, green/purple); class 0 slowest ($c \approx 0.750$, blue). The 4 : 2 : 1 floor ratio at $L = 12$ is the principal result (Table 3).

genuine geodesic, not a near-trivial element. It belongs to homology class 4 (non-identity coset) despite having nearly vanishing twist.

5 Discussion

The main findings are:

1. All five homology classes of $\pi_1(\mathbf{m006})$ exhibit exponential decay of their spectral floors with word length, with class-dependent rates c_k .
2. The rates form a $4 : 2 : 1$ hierarchy: $c_{1,4} \approx 0.985$, $c_{2,3} \approx 0.401$, $c_0 \approx 0.750$, confirmed by AIC over lengths 1–12.
3. The pairing $c_1 = c_4$, $c_2 = c_3$ reflects the involution $k \mapsto -k \pmod{5}$ on $\mathbb{Z}/5 \cong H_1(\mathbf{m006})$. Class 0 (identity coset) is self-paired and has an intermediate rate, not the fastest or slowest.
4. At length 12, the Meyerhoff manifold $\mathbf{m003}$ (also $H_1 = \mathbb{Z}/5$) exhibits the *same* pairing structure as $\mathbf{m006}$: floors $\phi^{(1,4)} = 0.04319^\circ$, $\phi^{(2,3)} = 0.09387^\circ$, $\phi^{(0)} = 0.04373^\circ$, giving ratio $\phi^{(2,3)}/\phi^{(1,4)} \approx 2.17$. The pairing $c_1 = c_4$, $c_2 = c_3$ thus appears to be a property of the $\mathbb{Z}/5$ torsion structure shared by both manifolds, not specific to $\mathbf{m006}$'s arithmetic.

5.1 Floors vanish: consequence of the Ambient PGT

Proposition 1. *Let M be a compact hyperbolic 3-manifold with $H_1(M, \mathbb{Z}) = \mathbb{Z}/5$. For each homology class $k \in \mathbb{Z}/5$, $\phi_{\text{floor}}^{(k)}(L) \rightarrow 0$ as $L \rightarrow \infty$.*

Proof. By the Ambient Prime Geodesic Theorem [4] and the equidistribution of homology classes [6], for any $\epsilon > 0$ the count of class- k geodesics with $\phi_{\text{fold}} < \epsilon$ and length $\leq T$ grows as $(4\epsilon/\pi) \cdot e^{2T}/(10T) \rightarrow \infty$. Hence every class eventually contains geodesics with $\phi_{\text{fold}} < \epsilon$, so $\phi_{\text{floor}}^{(k)}(L) \rightarrow 0$. $\square$

Remark 1. *Proposition 1 shows convergence to zero but says nothing about the rate. The class-dependent rates c_k (Table 4) are the new content of this paper.*

5.2 Margulis asymptotics and the exponential decay law

Proposition 1 guarantees $\phi_{\text{floor}}^{(k)}(L) \rightarrow 0$ but does not predict the rate. Here we explain why exponential decay is the natural expectation from the prime geodesic theorem.

The prime geodesic theorem (Margulis [5]) gives

$$\pi_{\Gamma}(T) \sim \frac{e^{2T}}{2T}, \quad T \rightarrow \infty, \quad (4)$$

for the count of primitive closed geodesics of length $\leq T$. The Ambient PGT [4] and the equidistribution of homology classes [6] together give, for each class $k \in \mathbb{Z}/5$ and each arc $J_{\epsilon} = \{\theta : |\phi_{\text{fold}}| < \epsilon\}$ of measure $|J_{\epsilon}| = 4\epsilon$,

$$N^{(k)}(T, \epsilon) := \#\{\gamma \in C_k : \ell(\gamma) \leq T, \phi_{\text{fold}}(\gamma) < \epsilon\} \sim \frac{4\epsilon}{\pi} \cdot \frac{1}{5} \cdot \frac{e^{2T}}{2T}. \quad (5)$$

The spectral floor $\phi_{\text{floor}}^{(k)}(L)$ is the value of ϵ at which $N^{(k)}(T_L, \epsilon)$ first reaches 1, where T_L is the maximum geodesic length achievable at word length L . Since each generator has a positive translation length bounded below by the systole $\text{sys}(\mathfrak{m}006) > 0$, the translation length of a word of length L is bounded above by $L \cdot \ell_{\text{max}}$ and below by ℓ_{min} , giving $T_L \leq L \cdot \ell_{\text{max}}$; empirically $T_L \sim \alpha L$ with $\alpha \approx \ell_{\text{max}}(\mathfrak{m}006) \approx 1.5$ for the word lengths studied. Since $T_L \sim \alpha L$ for some constant $\alpha > 0$ determined by the generator lengths, equation (5) gives

$$\phi_{\text{floor}}^{(k)}(L) \sim C_k \cdot e^{-2\alpha L}, \quad (6)$$

where the prefactor C_k encodes the leading-order correction to the equidistribution of twist angles within class k . The Margulis asymptotics predict that the *leading rate* 2α is the same for all classes, but the *prefactor* C_k can differ. The class-dependent rates c_k observed in our data (Table 4) are therefore corrections to the leading term — they reflect class-to-class variation in the density of near-zero-twist geodesics beyond what equidistribution predicts.

The pairing $c_1 = c_4, c_2 = c_3$ (Table 4, Figures 1) is then a statement about the symmetry of these corrections under the involution $k \mapsto -k \pmod{5}$. We conjecture that this symmetry is arithmetic in origin.

Conjecture 1. *The decay rates c_k are determined by the arithmetic invariants of $\pi_1(\mathfrak{m}006)$. Specifically, the pairing $c_k = c_{5-k}$ for $k \neq 0$ reflects an outer involution $\rho \mapsto \bar{\rho}$ of the holonomy representation $\rho : \pi_1(\mathfrak{m}006) \rightarrow \text{PSL}(2, \mathbb{C})$, arising from complex conjugation in the invariant trace field. Manifolds with the same invariant trace field and quaternion algebra as $\mathfrak{m}006$ have the same decay rate hierarchy $(c_0, c_1, c_2, c_3, c_4)$.*

A proof of this conjecture would require computing the invariant trace field of $\mathfrak{m}006$ via the SNAP package [10] and relating the arithmetic involution to the observed rate pairing. A direct test is feasible: SNAP computes the invariant trace field $k(M)$ and quaternion algebra $\mathcal{A}$ of any cusped hyperbolic

manifold. Running SNAP on the Dehn-filling description of `m006` and checking whether $\mathcal{A}$ admits a complex-conjugation involution would either confirm or refute the arithmetic origin of the pairing $c_1 = c_4$, $c_2 = c_3$. We note that `m003` is proven arithmetic (invariant trace field of discriminant -283) [9]; the arithmeticity of `m006` is expected but unproven.

The exponential decay rates c_k provide a new family of geometric invariants of compact hyperbolic 3-manifolds with torsion in H_1 . Comparing these rates across the OrientableClosedCensus for other manifolds with $H_1 = \mathbb{Z}/p$ (p prime) would test the conjecture and may reveal which arithmetic properties control the rate hierarchy.

Remark 2 (Scope of the decay claim). *The exponential fits in Table 4 and Figure 1 are empirical descriptions of the data over the range $L = 1, \dots, 12$. No claim is made regarding the asymptotic behaviour of $\phi_{\text{floor}}^{(k)}(L)$ as $L \rightarrow \infty$ beyond the convergence to zero established in Proposition 1. The Margulis asymptotics (Section 5.2) suggest exponential decay is natural, but a rigorous proof of the rate is an open problem.*

We note that the word-length L depends on the choice of generators $\{a, b\}$ for $\pi_1(\text{m006})$. A generator-independent formulation would use the geodesic translation length $\ell(\gamma) = 2 \log |\lambda(\gamma)|$ in place of L . A preliminary analysis using the running maximum of $\ell(\gamma)$ as the x -axis variable confirms that the class pairing $c_1 = c_4$ and $c_2 = c_3$ persists, though the rate values differ; a rigorous treatment via the geodesic length spectrum of `m006` is deferred to future work.

Data Availability

All census data and analysis scripts supporting this study are publicly available at <https://github.com/drmlgentry/hyperbolic-flavor-scan>. The complete word-length-12 census files (`twist_census_len12_m003.csv` and `twist_census_len12_m006.csv`) are included in the repository.

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Conflict of Interest

The author declares no conflicts of interest.

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